

Secretary Problem Research

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Lecture 3: Relaxed SP with error

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1 Calculation with error involved

Now we define the game and strategy as followed.

The weights $X_1, \dots, X_n \sim \text{Unif}(0, 1)$. The noisy observations $\tilde{X}_1, \dots, \tilde{X}_n \sim \text{Unif}(x_i - \delta, x_i + \delta)$.

The strategy is to introduce thresholding parameters R and $T \in (0, 1)$. We examine the upcoming item i if $\tilde{x}_i > R$, and accept the item if $x_i > T$. We still want to find $P[X_{max} is selected]$ given particular R, T, n, δ .

$$P[X_{max} \text{ selected}] = \sum_i \int_0^1 P[X_i \text{ best} \cap i \text{ selected} \mid X_i = x] f(x) dx \quad (1)$$

$$P[X_i \text{ best} \cap i \text{ selected} \mid X_i = x] = P[\text{reach } i \cap \tilde{X}_i > R \cap X_i > T \cap X_i \text{ max} \mid X_i = x] \quad (2)$$